Theory and Implementation of Shape Sensitivities in a Rapid Hypersonic Impact Method (HI-Mach)

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This paper describes a method for obtaining sensitivities of a quantity of interest with respect to the geometry in a hypersonic panel code. The hypersonic panel code (HI-Mach) implements impact methods to approximate quantities such as lift, drag, wall temperature, and heat flux. Because many of the methods within the hypersonic impact method are simple, direct analytical derivatives of these methods are obtained and presented. In the few cases where the analytic approach involves exceptionally complex derivatives, approximate techniques are described. The derivative calculations were verified against finite-difference results and showed near machine-precision agreement. The computational complexity is $O(n^2)$ with respect to the mesh density in terms of the number of vertices. Coupling with Engineering Sketch Pad (ESP) expresses the sensitivities with respect to vehicle design parameters. Sensitivities of quantities of interest with respect to the design parameters allow for efficient and fast optimization in the design process. Rapid design iteration is enabled by the direct sensitivity method described in this paper executing two orders of magnitude faster than finite differencing.

Nomenclature

A	=	panel area
a, b, c	=	velocity distribution constants
c_f	=	skin friction coefficient
C_p	=	panel pressure coefficient
C_L, C_D	=	lift, drag coefficient
$\boldsymbol{C_F}, \boldsymbol{C_M}$	=	force and moment coefficient
l	=	length
m	=	number of mesh panels
M	=	Mach number
n	=	number of mesh vertices
P	=	pressure
$\mathcal{P}$	=	CAD geometry design parameter
$\boldsymbol{p}$	=	vertex location vector
$\dot{q}$	=	heat flux
R	=	specific gas constant
Re	=	Reynolds number
S	=	Sutherland's law viscosity parameter
T	=	temperature
u	=	velocity
V_i	=	panel velocity vector
V_∞	=	freestream velocity vector

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$\hat{\boldsymbol{v}}_\infty$	=	freestream direction unit vector
x	=	total running length
x_i	=	single panel running length
$\hat{x}_i$	=	panel streamline end point
$\hat{x}_{i-1}$	=	panel streamline start point
$\boldsymbol{e}$	=	panel edge vector
$\hat{\boldsymbol{n}}$	=	panel unit normal vector
$\hat{\boldsymbol{a}}_{\boldsymbol{L}}$	=	attachment line direction vector
$\boldsymbol{\chi}$	=	discretized mesh represented as a column vector of all vertex coordinates
β	=	shock angle
ϵ	=	emissivity
γ	=	specific heat ratio
Λ	=	sweep angle
λ	=	velocity eigenvalue
ξ, η	=	velocity eigenvectors
μ	=	dynamic viscosity
ν	=	Prandtl-Meyer function
ϕ, ψ	=	parameters for calculating β
σ	=	Stefan-Boltzmann constant
τ	=	shear stress
θ	=	panel inclination angle relative to freestream
Subscripts		
2	=	post-shock property
aw	=	adiabatic wall property
e	=	boundary layer edge property
i, j, k	=	vertex or panel indices
attach, perp	=	denote curvature properties in the attachment line direction and direction perpendicular to that
s	=	stagnation property
w	=	wall property
∞	=	freestream property
Superscripts		
$*$	=	thermodynamic property evaluated at reference temperature

I. Introduction

The difficulties of the hypersonic flight environment require integrated and optimized designs. A successful hypersonic vehicle must handle high heat fluxes and extreme aerodynamic loads while maintaining performance. Thermal protection systems, control surfaces, propulsion elements, and other components greatly affect each other. Therefore, optimizing a complete vehicle requires many iterations of design and analysis on the entire vehicle. This multidisciplinary analysis and optimization (MDAO) requires relating changes in vehicle design to changes in performance.

Due to the difficulty associated with hypersonic optimization, previous efforts have focused on parameterized geometries with a small number of parameters, or specific vehicle components. In particular, inlets and forebodies have been studied extensively, because vehicle design and performance are highly sensitive to these components [1–3]. Other approaches to hypersonic geometry optimization have used gradient-based optimizers on a parameterized geometry with a small number of parameters [4].

An analysis approach with lower fidelity but higher computational speed allows for design and optimization at a faster pace. Once design and optimization is completed, higher fidelity models including Navier-Stokes codes can be used to validate the solution produced by a lower-fidelity model. For example, inviscid Euler methods have been successfully applied to speed up the hypersonic optimization process [5]. Another approach uses independent panel impact methods. The nature of hypersonic flow allows for these impact methods to produce reasonable aerothermodynamic results on unstructured meshes. The authors of this work developed a Hypersonic Impact Method for Aerodynamics and Convective Heating (HI-Mach) [6, 7]. Another common code which uses impact methods is CBAero [8], which has previously been used for optimization of hypersonic vehicles [9].

This work covers the derivation and calculation of geometric shape sensitivities in HI-Mach, which is an unstructured impact method for calculating aerodynamics and convective heating at hypersonic conditions. The mathematics governing the sensitivities of desired quantities are derived. The implementation of shape sensitivities is described, including discussion about the strengths and weaknesses of the methods. Results comparing the accuracy and computational speed of the shape sensitivities are presented.

II. Theory

In a design and optimization process, the gradient or sensitivity of a quantity is frequently desired. The sensitivity of a desired functional such as lift or drag is generally expressed with respect to some set of parameters. For example, in optimization of a parametrized airfoil shape, the sensitivity of lift-to-drag ratio with respect to camber and thickness distribution guides the optimizer to the best camber and thickness distribution. Previously, sensitivities were computed for lifting-line theory with Dual Number Automatic Differentiation [10]. Also, sensitivities were computed with the adjoint method for a subsonic and supersonic panel method [11].

Generally, an analysis code with sensitivity capability outputs the sensitivity of a functional with respect to all the points on a discretized geometry. A Computer-Aided Design tool such as Engineering Sketch Pad converts the sensitivity with respect to the geometry to the sensitivity with respect to the parameters [12].

Multiple approaches are commonly used to compute the sensitivities of a functional with respect to the geometry in analysis codes. Detailed discussions of these sensitivity calculation methods are given by Peter and Dwight [13] and Martins and Hwang [14]. A brief review is given here. The simplest approach is using finite-differencing to approximate the sensitivity. An initial solution is obtained, the inputs are perturbed, and a new solution is obtained. The estimate of the sensitivity is obtained as a function of the change in the solution and perturbation size. For example, in central-differencing, the sensitivity is estimated as:

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h} \tag{1}$$

where h is a small step size added to or subtracted from the independent (design) variable(s) x. While the central-difference method is easy to implement, it requires two function evaluations for each derivative estimate. This may be acceptable when calculating the sensitivities with respect to a few parameters. However, calculating the shape sensitivities of a geometry requires calculating the sensitivity of each vertex in each of the three coordinate directions. Therefore, calculating the shape sensitivity using central-differencing requires evaluating the solution $6n$ times, where n is the number of vertices. This is prohibitively expensive for large geometries. The central-difference method also has limited accuracy due to round-off and truncation error. With very small step sizes, subtractive cancellation or round-off error reduces the accuracy of the method. For large step sizes, truncation error dominates. The trade-off between round-off and truncation error is described by Kenway [15].

An alternative to traditional finite-difference methods is the complex-step approximation. The general form is

$$f'(x) \approx \text{imag}\left(\frac{f(x+hi)}{h}\right) \tag{2}$$

A derivation of the complex-step approximation is given by Martins et al. [16]. Since only a forward step is taken in the imaginary direction, the complex-step method requires $3n$ runs compared to the $6n$ required by central difference. While the complex-step method can be faster and more accurate than a finite-difference method, the number of required solution evaluations is prohibitive.

Another approach is automatic differentiation. Software packages can automatically produce code which runs alongside a program and calculates the gradients as the program executes. This approach has been successfully used in the development of discrete adjoint Navier-Stokes codes [17]. However, automatic differentiation implementations can suffer from memory inefficiencies. In addition, a bespoke differentiation approach where derivatives can be easily computed can outperform an automatic differentiation approach.

Many codes have been differentiated using the adjoint method. The adjoint method uses a linear algebra technique to obtain gradients of the solution alongside the solution. The adjoint method has been successfully applied to Navier-Stokes equations [18]. However, the computational methods used within HI-Mach are much simpler than a Navier-Stokes code. No large linear or non-linear systems are solved, nor are any PDEs solved. The adjoint method is therefore not appropriate for differentiating HI-Mach. HI-Mach is instead directly differentiated to obtain the sensitivities.

A. Definitions

HI-Mach runs unstructured meshes composed of triangular panels [6]. These meshes are a discrete representation of a geometry. Mathematically, a mesh of a geometry is composed of n vertices, and is described by the set of vertex coordinates

$$\boldsymbol{\chi} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \\ y_1 \\ \vdots \\ y_n \\ z_1 \\ \vdots \\ z_n \end{bmatrix} \tag{3}$$

which is a $3n \times 1$ column vector. Vertices within a geometry are connected together into m panels.

Vertex position is described by the 3×1 column vector

$$\mathbf{p}_i = \begin{bmatrix} x_i \\ y_i \\ z_i \end{bmatrix} \tag{4}$$

The edge going from vertex i to vertex j is

$$\mathbf{e}_{ij} = \mathbf{p}_j - \mathbf{p}_i \tag{5}$$

which is also a 3×1 column vector.

In this work, the derivative of a vector dot product or vector cross product is computed. For a dot product, the derivative is of the form

$$\frac{\partial(\boldsymbol{a} \cdot \boldsymbol{b})}{\partial \boldsymbol{\chi}} = \frac{\partial \boldsymbol{a}}{\partial \boldsymbol{\chi}} \cdot \boldsymbol{b} + \boldsymbol{a} \cdot \frac{\partial \boldsymbol{b}}{\partial \boldsymbol{\chi}} \tag{6}$$

The terms on the right hand side are technically Jacobian-vector products, not vector dot products. Taking the derivative of $\boldsymbol{a}$ or $\boldsymbol{b}$ with respect to $\boldsymbol{\chi}$, a vector, results in a Jacobian. However, in this paper these derivative matrix-vector products are left in dot product form. Similarly, for a cross product the derivative is of the form

$$\frac{\partial(\boldsymbol{a} \times \boldsymbol{b})}{\partial \boldsymbol{\chi}} = \frac{\partial \boldsymbol{a}}{\partial \boldsymbol{\chi}} \times \boldsymbol{b} + \boldsymbol{a} \times \frac{\partial \boldsymbol{b}}{\partial \boldsymbol{\chi}} \tag{7}$$

The right hand side terms are technically matrix products, since taking the cross product of a matrix with a vector is not well-defined. Functionally, this operation is equivalent to row-wise cross products. The notation is left unmodified throughout this work for convenience.

B. Geometric Derivatives

The derivative of a vertex position with respect to the set of vertex coordinates is a $3 \times n$ matrix. For the i-th vertex position, the derivative is

$$\frac{\partial \mathbf{p}_i}{\partial \boldsymbol{\chi}} = \begin{bmatrix} 0 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 1 & 0 & 0 \\ 0 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 0 & 0 & 1 \end{bmatrix}^{\boldsymbol{T}} \tag{8}$$

The derivative of the edge with respect to the coordinate vector is

$$\frac{\partial \mathbf{e}_{ij}}{\partial \boldsymbol{\chi}} = \frac{\partial \mathbf{p}_j}{\partial \boldsymbol{\chi}} - \frac{\partial \mathbf{p}_i}{\partial \boldsymbol{\chi}} \tag{9}$$

which is also a $3 \times n$ matrix. It is useful to define the edge normal vector

$$\hat{\mathbf{e}}_{ij} = \frac{\mathbf{e}_{ij}}{|\mathbf{e}_{ij}|} \tag{10}$$

The derivative of the edge vector magnitude is

$$\frac{\partial |\mathbf{e}_{ij}|}{\partial \boldsymbol{\chi}} = \frac{1}{2|\mathbf{e}_{ij}|} \frac{\partial}{\partial \boldsymbol{\chi}} \left(\mathbf{e}_{ij} \cdot \mathbf{e}_{ij} \right) = \frac{1}{|\mathbf{e}_{ij}|} \left(\frac{\partial \mathbf{e}_{ij}}{\partial \boldsymbol{\chi}} \right) \mathbf{e}_{ij} = \left(\frac{\partial \mathbf{e}_{ij}}{\partial \boldsymbol{\chi}} \right) \cdot \hat{\mathbf{e}}_{ij} \tag{11}$$

Using this definition, the derivative of the edge normal vector can be expressed as

$$\frac{\partial \hat{\mathbf{e}}_{ij}}{\partial \boldsymbol{\chi}} = \frac{|\mathbf{e}_{ij}| \frac{\partial \mathbf{e}_{ij}}{\partial \boldsymbol{\chi}} - \frac{\partial |\mathbf{e}_{ij}|}{\partial \boldsymbol{\chi}} \mathbf{e}_{ij}}{\mathbf{e}_{ij} \cdot \mathbf{e}_{ij}} \tag{12}$$

which is a $3 \times n$ matrix.

A panel unit normal vector is defined by [6]

$$\hat{\mathbf{n}} = \frac{\mathbf{e}_{ij} \times \mathbf{e}_{jk}}{|\mathbf{e}_{ij} \times \mathbf{e}_{jk}|} \tag{13}$$

The derivative of the panel unit normal with respect to the coordinates is then obtained as

$$\frac{\partial \hat{\mathbf{n}}}{\partial \boldsymbol{\chi}} = \frac{|\mathbf{e}_{ij} \times \mathbf{e}_{jk}| \left[\left(\mathbf{e}_{ij} \times \frac{\partial \mathbf{e}_{jk}}{\partial \boldsymbol{\chi}} \right) + \left(\frac{\partial \mathbf{e}_{ij}}{\partial \boldsymbol{\chi}} \times \mathbf{e}_{jk} \right) \right] - \frac{\partial}{\partial \boldsymbol{\chi}} \left(|\mathbf{e}_{ij} \times \mathbf{e}_{jk}| \right) \mathbf{e}_{ij} \times \mathbf{e}_{jk}}{\left(\mathbf{e}_{ij} \times \mathbf{e}_{jk} \right) \cdot \left(\mathbf{e}_{ij} \times \mathbf{e}_{jk} \right)} \tag{14}$$

where

$$\frac{\partial}{\partial \boldsymbol{\chi}} \left(|\mathbf{e}_{ij} \times \mathbf{e}_{jk}| \right) = \left[\left(\mathbf{e}_{ij} \times \frac{\partial \boldsymbol{e}_{jk}}{\partial \boldsymbol{\chi}} \right) + \left(\frac{\partial \mathbf{e}_{ij}}{\partial \boldsymbol{\chi}} \times \mathbf{e}_{jk} \right) \right] \cdot \frac{\mathbf{e}_{ij} \times \mathbf{e}_{jk}}{|\mathbf{e}_{ij} \times \mathbf{e}_{jk}|} \tag{15}$$

This is also a $3 \times n$ matrix.

The panel inclination relative to the freestream is given by [6]

$$\theta = \cos^{-1} \left(\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{v}}_\infty \right) - \frac{\pi}{2} \tag{16}$$

where $\hat{\boldsymbol{v}}_\infty$ is the normalized freestream velocity direction. Therefore, the derivative of a panel normal vector with respect to the mesh vertices is

$$\frac{\partial \theta}{\partial \boldsymbol{\chi}} = -\frac{1}{\sqrt{1-(\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{v}}_\infty)^2}}\frac{\partial \hat{\boldsymbol{n}}}{\partial \boldsymbol{\chi}}\cdot\hat{\boldsymbol{v}}_\infty \tag{17}$$

This is a $1 \times n$ row vector. Therefore, the complete set of derivatives of all of the panel inclinations with respect to all of the vertex locations is an $m \times n$ matrix.

The area of the panel made up of vertices $\boldsymbol{p}_i$, $\boldsymbol{p}_j$, and $\boldsymbol{p}_k$ is given by

$$A = \frac{|\boldsymbol{e}_{ij}\times\boldsymbol{e}_{jk}|}{2} = \frac{\sqrt{(\boldsymbol{e}_{ij}\times\boldsymbol{e}_{jk})\cdot(\boldsymbol{e}_{ij}\times\boldsymbol{e}_{jk})}}{2} \tag{18}$$

Therefore, the derivative of panel area with respect to the geometry is

$$\frac{\partial A}{\partial \boldsymbol{\chi}} = \frac{1}{2\sqrt{(\boldsymbol{e}_{ij}\times\boldsymbol{e}_{jk})\cdot(\boldsymbol{e}_{ij}\times\boldsymbol{e}_{jk})}}\left(\frac{\partial}{\partial \boldsymbol{\chi}}\left[\boldsymbol{e}_{ij}\times\boldsymbol{e}_{jk}\right]\right)\cdot(\boldsymbol{e}_{ij}\times\boldsymbol{e}_{jk}) \tag{19}$$

which is a $1 \times n$ row vector.

C. Inviscid Pressure Derivatives

The core of HI-Mach's aerodynamic methods is an inviscid pressure solution. The approach to the inviscid pressure solution is described in detail in [6]. In short, the pressure on each panel is considered independent of the surrounding panels. In addition, the pressure is considered to be purely a function of the panel inclination with respect to the freestream, θ.

Panels on a geometry are classified as one of three types:

- Windward panels are inclined towards the freestream: $\theta > 0$
- Leeward panels are inclined away from the freestream, but not inclined beyond the maximum flow turning angle: $\theta_{\min} < \theta < 0$
- Base panels are inclined far enough away from the freestream that the flow is assumed to have separated: $\theta < \theta_{\min}$

The pressure is treated differently for windward, leeward, and base panels.

The pressure on base panels is a constant, so it does not vary with panel inclination. Thus, the derivative of the pressure on base panels is 0. The pressure on windward and leeward panels varies with respect to panel inclination. The derivative of pressure on windward & leeward panels is described below.

1. Windward Methods

Various methods are available to calculate the pressure in a windward region [6]. Both the straight and modified Newtonian methods take the form

$$C_{p,\text{Newtonian}} = C_{p,\max}\sin^2(\theta) \tag{20}$$

Therefore, the derivative of a panel pressure with respect to the vertex coordinates is

$$\frac{\partial C_{p,\text{Newtonian}}}{\partial \boldsymbol{\chi}} = 2C_{p,\max}\sin(\theta)\cos(\theta)\frac{\partial \theta}{\partial \boldsymbol{\chi}} = C_{p,\max}\sin(2\theta)\frac{\partial \theta}{\partial \boldsymbol{\chi}} \tag{21}$$

The tangent-cone method is also implemented in HI-Mach as a Windward pressure method. For computational speed, it is implemented using a tabulated database of coefficients such that

$$C_{p,\text{tc}} = C_0 + C_1\theta + C_2\theta^2 + C_3\theta^3 + C_4\theta^4 + C_5\theta^5 + C_6\theta^6 \tag{22}$$

The derivative of the tangent cone panel pressure coefficient with respect to the vertex locations is then trivially found

$$\frac{\partial C_{p,\text{tc}}}{\partial \boldsymbol{\chi}} = \left(C_1 + 2C_2\theta + 3C_3\theta^2 + 4C_4\theta^3 + 5C_5\theta^4 + 6C_6\theta^5\right)\frac{\partial \theta}{\partial \boldsymbol{\chi}} \tag{23}$$

The tangent-wedge method is not reproduced here for brevity but is covered in Eqs. (5)–(9) of [6]. The derivative of the tangent-wedge pressure coefficient with respect to the vertex coordinates is obtained using the chain rule

$$\frac{\partial C_{p,\mathrm{tw}}}{\partial \chi} = \frac{2}{\gamma M_\infty^2 P_\infty}\frac{\partial P_2}{\partial \chi} = \frac{4}{\gamma+1}\sin(2\beta)\frac{\partial \beta}{\partial \chi} \tag{24}$$

The $\partial\beta/\partial\chi$ term is difficult to obtain. The chain rule gives

$$\frac{\partial \beta}{\partial \chi} = \frac{\partial \beta}{\partial \theta}\frac{\partial \theta}{\partial \chi} \tag{25}$$

where

$$\frac{\partial \beta}{\partial \theta} = \frac{\dfrac{2\cos\left(\frac{4\pi}{3}+\frac{\mathrm{acos}(\phi)}{3}\right)\frac{\partial}{\partial\theta}\psi + \frac{2\sin\left(\frac{4\pi}{3}+\frac{\mathrm{acos}(\phi)}{3}\right)\psi\frac{\partial}{\partial\theta}\phi}{3\sqrt{1-\phi^2}}}{\tan(\theta)\left(3M^2\left(\frac{\gamma-1}{2}\right)+3\right)} - \dfrac{\left(\tan(\theta)^2+1\right)\left(M^2+2\cos\left(\frac{4\pi}{3}+\frac{\mathrm{acos}(\phi)}{3}\right)\psi-1\right)}{\tan(\theta)^2\left(3M^2\left(\frac{\gamma-1}{2}\right)+3\right)}}{\dfrac{\left(M^2+2\cos\left(\frac{4\pi}{3}+\frac{\mathrm{acos}(\phi)}{3}\right)\psi-1\right)^2}{\tan(\theta)^2\left(3M^2\left(\frac{\gamma-1}{2}\right)+3\right)^2}+1} \tag{26}$$

The variables ψ and ϕ are defined as

$$\psi = \sqrt{(M^2-1)^2 - 3\left(1+\frac{\gamma-1}{2}M^2\right)\left(1+\frac{\gamma+1}{2}M^2\right)\tan^2\theta} \tag{27}$$

and

$$\phi = \frac{(M^2-1)^3 - 9\left(1+\frac{\gamma-1}{2}M^2\right)\left(1+\frac{\gamma-1}{2}M^2+\frac{\gamma+1}{4}M^4\right)\tan^2\theta}{\psi^3} \tag{28}$$

and their partial derivatives with respect to θ are

$$\frac{\partial \psi}{\partial \theta} = -\frac{\tan(\theta)\left(\left(\frac{\gamma+1}{2}\right)M^2+1\right)\left(3M^2\left(\frac{\gamma-1}{2}\right)+3\right)\left(\tan(\theta)^2+1\right)}{\sqrt{\left(M^2-1\right)^2 - \tan(\theta)^2\left(\left(\frac{\gamma+1}{2}\right)M^2+1\right)\left(3M^2\left(\frac{\gamma-1}{2}\right)+3\right)}} \tag{29}$$

and

$$\frac{\partial \phi}{\partial \theta} = -\frac{3\left(\left(M^2-1\right)^3 - \tan^2\theta\left(9M^2\left(\frac{\gamma-1}{2}\right)+9\right)\left(\left(\frac{\gamma+1}{4}\right)M^4+\left(\frac{\gamma-1}{2}\right)M^2+1\right)\right)\frac{\partial\psi}{\partial\theta}}{\psi^4} - \frac{2\tan\theta\left(9M^2\left(\frac{\gamma-1}{2}\right)+9\right)\left(\tan^2\theta+1\right)\left(\left(\frac{\gamma+1}{4}\right)M^4+\left(\frac{\gamma-1}{2}\right)M^2+1\right)}{\psi^3}.$$

The blended tangent wedge-tangent cone method is used for analysis of conically derived waveriders. The pressure coefficient is a weighted average of the tangent wedge and tangent cone methods, with the weighting determined by the local surface inclination. The pressure coefficient is given by

$$C_{p,\mathrm{blended}} = \left[1 - \frac{(\delta-\theta)^2}{(\delta-\theta_c)^2}\right]C_{p,\mathrm{tw}} + \frac{(\delta-\theta)^2}{(\delta-\theta_c)^2}C_{p,\mathrm{tc}} \tag{30}$$

where δ and θ_c are problem constants. Therefore, the derivative of the blended pressure is equal to

$$\frac{\partial C_{p,\mathrm{blended}}}{\partial \chi} = \frac{\partial C_{p,\mathrm{blended}}}{\partial \theta}\frac{\partial \theta}{\partial \chi} = \left\{\frac{\partial C_{p,\mathrm{tw}}}{\partial \theta} - 2\left[\frac{\delta-\theta}{(\delta-\theta_c)^2}\right]\left(C_{p,\mathrm{tc}} - C_{p,\mathrm{tw}}\right) + \left[\frac{(\delta-\theta)^2}{(\delta-\theta_c)^2}\right]\left(\frac{\partial C_{p,\mathrm{tc}}}{\partial \theta} - \frac{\partial C_{p,\mathrm{tw}}}{\partial \theta}\right)\right\}\frac{\partial \theta}{\partial \chi} \tag{31}$$

2. Leeward derivatives

The pressure coefficient on expansion surfaces is given by the Prandtl-Meyer relation, which determines the boundary layer edge Mach number on an expansion surface [6]. The Mach number is then converted to pressure using the isentropic relations. The panel pressure coefficient is given by

$$C_{p,\text{expansion}} = \frac{2}{\gamma M_\infty^2}\left[\left(\frac{1+\frac{\gamma-1}{2}M_\infty^2}{1+\frac{\gamma-1}{2}M^2}\right)^{\frac{\gamma}{\gamma-1}} - 1\right] \tag{32}$$

The derivative of the pressure coefficient with respect to the geometry can be expanded using the chain rule

$$\frac{\partial C_{p,\text{expansion}}}{\partial \chi} = \frac{\partial C_{p,\text{expansion}}}{\partial M}\frac{\partial M}{\partial \theta}\frac{\partial \theta}{\partial \chi} \tag{33}$$

The first term in the chain rule sequence is obtained by application of derivative rules

$$\frac{\partial C_{p,\text{expansion}}}{\partial M} = -\frac{2M}{M_\infty^2}\left(\frac{1+\frac{\gamma-1}{2}M_\infty^2}{1+\frac{\gamma-1}{2}M^2}\right)^{\frac{1}{\gamma-1}}\left(\frac{1+\frac{\gamma-1}{2}M_\infty^2}{\left(1+\frac{\gamma-1}{2}M^2\right)^2}\right) \tag{34}$$

The second term is more difficult. The Mach Number relation is governed by the Prandtl-Meyer relation

$$\theta = \nu(M) - \nu(M_\infty) \tag{35}$$

Therefore, the Mach Number on the expansion surface can be expressed as

$$M = \nu^{-1}(\theta + \nu(M_\infty)) \tag{36}$$

Where $\nu^{-1}(\theta)$ denotes the inverse Prandtl-Meyer function, which is impossible to express in terms of elementary functions. Nevertheless, by differentiating we arrive at

$$\frac{\partial M}{\partial \theta} = \frac{\partial}{\partial \theta}\nu^{-1}(\theta + \nu(M_\infty)) \tag{37}$$

Which can be expressed using the inverse derivative rule as

$$\frac{\partial M}{\partial \theta} = \frac{1}{\nu'(\nu^{-1}(\theta + \nu(M_\infty)))} \tag{38}$$

Where the derivative of the Prandtl-Meyer function is

$$\nu'(M) = \frac{\partial \nu(M)}{\partial M} = \frac{M\sqrt{\frac{\gamma+1}{\gamma-1}}\,(\gamma-1)}{\sqrt{\frac{(M^2-1)\,(\gamma-1)}{\gamma+1}}\,\left(M^2\gamma - M^2 + 2\right)} - \frac{1}{M\sqrt{M^2-1}} \tag{39}$$

The expression for the derivative of the Mach number with respect to theta still involves the inverse Prandtl-Meyer function, which cannot be directly calculated. However, the quantity $\nu^{-1}(\theta + \nu(M_\infty))$ is the Mach number on the expansion surface, which is already computed [6]. In this implementation, the inverse Prandtl-Meyer function is evaluated based on a table lookup.

The other leeward pressure methods are for shielded and base panels. None of these have any sensitivity in the pressure with respect to the geometry.

D. Thermodynamic Property derivatives

HI-Mach produces estimates of viscous drag and convective heating [6, 7]. These estimates require the calculation of the gas thermodynamic properties on each panel. Derivatives of these properties with respect to the geometry are produced here.

The pressure on a panel is given by Eq. (24) of [6]. The derivative with respect to the geometry is

$$\frac{\partial P_e}{\partial \chi} = \frac{P_\infty \gamma M_\infty^2}{2} \frac{\partial C_p}{\partial \chi} \tag{40}$$

where the second term has been determined previously. The pressure is assumed to be constant throughout the boundary layer.

The boundary layer edge Mach number is given by Eq. (25) of [6]. The derivative with respect to the geometry is

$$\frac{\partial M_e}{\partial \chi} = \frac{\partial M_e}{\partial P_e}\frac{\partial P_e}{\partial \chi} = \frac{-P_s}{\gamma P_e^2 \sqrt{\frac{2}{\gamma-1}\left(\left(\frac{P_s}{P_{e,i}}\right)^{\frac{\gamma-1}{\gamma}} - 1\right)}} \left(\frac{P_e}{P_s}\right)^{\frac{1}{\gamma}} \frac{\partial P_e}{\partial \chi} \tag{41}$$

The boundary layer edge temperature is defined by Eq. (30) of [6]. The derivative of the boundary layer edge temperature with respect to the geometry is

$$\frac{\partial T_e}{\partial \chi} = \frac{\partial T_e}{\partial M_e}\frac{\partial M_e}{\partial \chi} = \frac{-4T_s(\gamma-1)M_e}{\left(2+[\gamma-1]M_e^2\right)^2}\frac{\partial M_e}{\partial \chi} \tag{42}$$

The boundary layer edge density is also obtained through the ideal gas law. The derivative of the boundary layer edge density with respect to the geometry is

$$\frac{\partial \rho_e}{\partial \chi} = \frac{\partial \rho_e}{\partial P_e}\frac{\partial P_e}{\partial \chi} + \frac{\partial \rho_e}{\partial T_e}\frac{\partial T_e}{\partial \chi} = \frac{1}{RT_e}\frac{\partial P_e}{\partial \chi} - \frac{P_e}{RT_e^2}\frac{\partial T_e}{\partial \chi} \tag{43}$$

The dynamic viscosity is governed by Sutherland's law which is reproduced in Eq. (35) of [6]. The derivative of the dynamic viscosity with respect to geometry is

$$\frac{\partial \mu}{\partial \chi} = \mu_{\text{ref}}\left(\frac{3}{2}\sqrt{\frac{T}{T_{\text{ref}}}}\frac{T_{\text{ref}}+S}{T+S} - \left(\frac{T}{T_{\text{ref}}}\right)^{\frac{3}{2}}\frac{T_{\text{ref}}+S}{(T+S)^2}\right)\frac{\partial T}{\partial \chi} \tag{44}$$

S, μ_{ref}, and T_{ref} are reference properties for a particular gas. In this instance, the boundary layer edge temperature T_e is replaced with a general temperature, T, since the viscosity at the wall is also often considered.

In HI-Mach, the viscosity and Prandtl number can also be computed by referencing tabulated data with respect to temperature and pressure. The quantities are calculated by linear interpolation on the tabulated data, so the derivatives are trivially obtained.

1. Edge Velocities

Two quantities used for computing boundary layer edge velocity are $\hat{\boldsymbol{e}}_{\text{stream}}$ and $\hat{\boldsymbol{e}}_{\text{perp}}$. These unit vectors are aligned and perpendicular with the velocity on the surface of the panel, respectively. Their definitions and derivatives are:

$$\hat{\boldsymbol{e}}_{\text{stream}} = \frac{\boldsymbol{e}_{\text{stream}}}{|\boldsymbol{e}_{\text{stream}}|} = \frac{(\hat{\boldsymbol{n}} \times \boldsymbol{v}_\infty) \times \hat{\boldsymbol{n}}}{|(\hat{\boldsymbol{n}} \times \boldsymbol{v}_\infty) \times \hat{\boldsymbol{n}}|} \tag{45}$$

$$\frac{\partial \hat{\boldsymbol{e}}_{\text{stream}}}{\partial \chi} = \frac{\frac{\partial \boldsymbol{e}_{\text{stream}}}{\partial \chi}|\boldsymbol{e}_{\text{stream}}| - \boldsymbol{e}_{\text{stream}}\frac{\boldsymbol{e}_{\text{stream}}\cdot\frac{\partial \boldsymbol{e}_{\text{stream}}}{\partial \chi}}{|\boldsymbol{e}_{\text{stream}}|}}{|\boldsymbol{e}_{\text{stream}}|^2} \tag{46}$$

where

$$\frac{\partial \boldsymbol{e}_{\text{stream}}}{\partial \chi} = \left(\frac{\partial \hat{\boldsymbol{n}}}{\partial \chi} \times \boldsymbol{v}_\infty\right) \times \hat{\boldsymbol{n}} + (\hat{\boldsymbol{n}} \times \boldsymbol{v}_\infty) \times \frac{\partial \hat{\boldsymbol{n}}}{\partial \chi} \tag{47}$$

The vector perpendicular to the panel and the local streamline direction is

$$\hat{\boldsymbol{e}}_{\text{perp}} = \frac{\boldsymbol{e}_{\text{perp}}}{|\boldsymbol{e}_{\text{perp}}|} = \frac{\hat{\boldsymbol{n}} \times \hat{\boldsymbol{e}}_{\text{stream}}}{|\hat{\boldsymbol{n}} \times \hat{\boldsymbol{e}}_{\text{stream}}|} \tag{48}$$

Its derivative is computed similarly to the derivative of $\hat{\boldsymbol{e}}_{\text{stream}}$ as presented in Eq. (46), except the quantity

$$\frac{\partial \boldsymbol{e}_{\text{perp}}}{\partial \chi} = \frac{\partial \hat{\boldsymbol{n}}}{\partial \chi} \times \hat{\boldsymbol{e}}_{\text{stream}} + \hat{\boldsymbol{n}} \times \frac{\partial \hat{\boldsymbol{e}}_{\text{stream}}}{\partial \chi} \tag{49}$$

is used.

The magnitude of the boundary layer edge velocity V_e is given by Eq. (33) of [6]. The velocity is defined on a per-panel basis

$$\boldsymbol{V}_e = \hat{\boldsymbol{e}}_{\text{stream}} V_e \tag{50}$$

The derivative of the boundary layer edge velocity vector on a panel with respect to the geometry is

$$\frac{\partial \boldsymbol{V}_e}{\partial \chi} = \frac{\partial \boldsymbol{V}_e}{\partial M_e}\frac{\partial M_e}{\partial \chi} + \frac{\partial \boldsymbol{V}_e}{\partial T_e}\frac{\partial T_e}{\partial \chi} + \frac{\partial \boldsymbol{V}_e}{\partial \hat{\boldsymbol{e}}_{\text{stream}}}\frac{\partial \hat{\boldsymbol{e}}_{\text{stream}}}{\partial \chi} = \hat{\boldsymbol{e}}_{\text{stream}}\sqrt{\gamma R T_e}\frac{\partial M_e}{\partial \chi} + \hat{\boldsymbol{e}}_{\text{stream}} M_e \frac{\gamma R}{2\sqrt{\gamma R T_e}}\frac{\partial T_e}{\partial \chi} + V_e \frac{\partial \hat{\boldsymbol{e}}_{\text{stream}}}{\partial \chi} \tag{51}$$

where R is the specific gas constant of the gas in question.

E. Reference Temperature

For viscous calculations in the acreage regions of the body, a reference temperature is used to evaluate the thermodynamic properties. The reference temperature can be expressed conveniently as

$$T^* = a_1 T_e + a_2 T_w + a_3 T_e M_e^2 \tag{52}$$

where the values of coefficients a_1, a_2, and a_3 are available in [6] for various reference temperature models. It is convenient to split up the derivative of the reference temperature into two parts, one which depends on the derivative of the wall temperature, and one that does not

$$\frac{\partial T^*}{\partial \chi} = \left.\frac{\partial T^*}{\partial \chi}\right|_{T_w} + \frac{\partial T^*}{\partial T_w}\frac{\partial T_w}{\partial \chi} \tag{53}$$

Note that $|_{T_w}$ implies $\partial T_w/\partial \chi = 0$. These parts are found from Eq. (52)

$$\left.\frac{\partial T^*}{\partial \chi}\right|_{\partial T_w/\partial \chi = 0} = a_1 \frac{\partial T_e}{\partial \chi} + a_3 \left(M_e^2 \frac{\partial T_e}{\partial \chi} + 2 M_e T_e \frac{\partial M_e}{\partial \chi} \right) \tag{54}$$

and

$$\frac{\partial T^*}{\partial T_w} = a_2 \tag{55}$$

F. Friction Coefficient

The friction coefficient is used in calculating the viscous drag and acreage heating [6]. The friction coefficient can be generally expressed as

$$c_f = k_{c_f} \text{Re}^{-m} \tag{56}$$

where the values of k_{c_f} and Re^{-m} differ depending on which correlation is used for friction coefficient and can be found in [6]. Similarly to how the derivative of the reference temperature is split in Eq. (53), the derivative of the friction coefficient is obtained in two parts. The first neglects changes in wall temperature.

$$\left.\frac{\partial c_f}{\partial \chi}\right|_{T_w} = c_f m \left[\mu^{*^{-1}} \left.\frac{\partial \mu^*}{\partial \chi}\right|_{T_w} - \rho^{*^{-1}} \left.\frac{\partial \rho^*}{\partial \chi}\right|_{T_w} - u_e^{-1}\frac{\partial u_e}{\partial \chi} - x^{-1}\frac{\partial x}{\partial \chi} \right] \tag{57}$$

and

$$\frac{\partial c_f}{\partial T_w} = c_f m \left[\mu^{*-1}\frac{\partial \mu^*}{\partial T^*}\frac{\partial T^*}{\partial T_w} - \rho^{*-1}\frac{\partial \rho^*}{\partial T^*}\frac{\partial T^*}{\partial T_w} \right] \tag{58}$$

G. Stagnation Heating Derivative

The heating to a stagnation region is given by [7]

$$\dot{q}_s = k_{\text{geom}} \left(h_{aw} - h_w\right) \text{Pr}_{wf}^{-0.6} \rho_w^{0.07} \mu_w^{0.07} \rho_e^{0.43} \mu_e^{0.43} \left(\frac{du_e}{dx}\right)^{0.5} \tag{59}$$

where k_{geom} is a geometric factor based on whether the panel is cylindrical or spherical. Like the reference temperature and friction coefficient, the derivative of stagnation heating is split into two parts. We split up the derivative of the stagnation heating into two parts, one which depends on the derivative of the wall temperature, and one that does not. The derivative neglecting the wall temperature changes is

$$\left.\frac{\partial \dot{q}_s}{\partial \chi}\right|_{T_w} = \dot{q}_s \left[0.43\rho_e^{-1}\frac{\partial \rho_e}{\partial \chi} + 0.43\mu_e^{-1}\frac{\partial \mu_e}{\partial \chi} + 0.5\left(\frac{du_e}{dx}\right)^{-1}\frac{\partial \frac{du_e}{dx}}{\partial \chi} + 0.07\rho_w^{-1}\frac{\partial \rho_w}{\partial p}\frac{\partial p}{\partial \chi} + 0.07\mu_w^{-1}\frac{\partial \mu_w}{\partial p}\frac{\partial p}{\partial \chi} - 0.6\text{Pr}_w^{-1}\frac{\partial \text{Pr}_w}{\partial p}\frac{\partial p}{\partial \chi}\right] \tag{60}$$

The derivative with respect to wall temperature is similarly obtained

$$\frac{\partial \dot{q}_s}{\partial T_w} = \dot{q}_s \left[-\left(h_{aw} - h_w\right)^{-1}\frac{\partial h_w}{\partial T_w} + 0.07\rho_w^{-1}\frac{\partial \rho_w}{\partial T_w} + 0.07\mu_w^{-1}\frac{\partial \mu_w}{\partial T_w} - 0.6\text{Pr}_w^{-1}\frac{\partial \text{Pr}_w}{\partial T_w}\right] \tag{61}$$

1. Boundary Layer Edge Velocity Gradient

The derivative of the boundary layer edge velocity gradient is the most difficult to compute of all these terms. As shown in [7], there are multiple approaches to calculating the boundary layer edge velocity gradient, and the derivative is different for each approach. The velocity gradient depends on surface curvature. The Olivier velocity gradient method is used as a default in HI-Mach. The derivative of the Olivier velocity gradient is

$$\frac{\partial}{\partial \chi}\left(\frac{du_e}{dx}\right) = \frac{u_\infty}{\rho_\infty u_\infty^2 \rho_s}\frac{1+\bar{\Delta}}{\bar{\Delta}}\left[-\frac{1}{R^2}\frac{\partial R}{\partial \chi}\left(P_s - P_2\right)\rho_2\right] \tag{62}$$

where the derivative of the radius is related to curvature derivatives through

$$\frac{\partial R}{\partial \chi} = -\frac{1}{\kappa_{\text{perp}}^2}\frac{\partial \kappa_{\text{perp}}}{\partial \chi} \tag{63}$$

The curvature in the direction perpendicular to the attachment line, κ_{perp}, is related to the principal curvatures through

$$\kappa_{\text{attach}} = \cos^2\theta\left(\kappa_1 - \kappa_2\right) + \kappa_2 \tag{64}$$

$$\kappa_{\text{perp}} = \kappa_1 + \kappa_2 - \kappa_{\text{attach}} \tag{65}$$

where θ is the angle between the attachment line direction and the first principal direction of curvature. The $\cos^2\theta$ term and its derivative are given by

$$\cos^2\theta = \left(\hat{\boldsymbol{\xi}}_1 \cdot \hat{\boldsymbol{a}}_L\right)^2; \quad \frac{\partial \cos^2\theta}{\partial \chi} = 2\left(\hat{\boldsymbol{\xi}}_1 \cdot \hat{\boldsymbol{a}}_L\right)\left(\frac{\partial \hat{\boldsymbol{\xi}}_1}{\partial \chi}\cdot \hat{\boldsymbol{a}}_L + \hat{\boldsymbol{\xi}}_1 \cdot \frac{\partial \hat{\boldsymbol{a}}_L}{\partial \chi}\right) \tag{66}$$

where the principal curvature direction $\hat{\boldsymbol{\xi}}_1$ and the attachment line direction $\hat{\boldsymbol{a}}_L$ are assumed to be unit vectors. The derivative of curvature in the attachment direction is then

$$\frac{\partial \kappa_{\text{attach}}}{\partial \chi} = \frac{\partial \cos^2\theta}{\partial \chi}\left(\kappa_1 - \kappa_2\right) + \cos^2\theta\left(\frac{\partial \kappa_1}{\partial \chi} - \frac{\partial \kappa_2}{\partial \chi}\right) + \frac{\partial \kappa_2}{\partial \chi} \tag{67}$$

which leads to an expression for the derivative of the curvature in the perpendicular direction

$$\frac{\partial \kappa_{\text{perp}}}{\partial \chi} = \frac{\partial \kappa_1}{\partial \chi} + \frac{\partial \kappa_2}{\partial \chi} - \frac{\partial \kappa_{\text{attach}}}{\partial \chi} \tag{68}$$

2. Geometry Constant

The geometric constant is given as

$$k_{\text{geom}} = 0.767\sqrt{\frac{1 + |\frac{\kappa_{\text{attach}}}{\kappa_{\text{perp}}}|}{2}} \tag{69}$$

where the ratio $\kappa_{\text{attach}}/\kappa_{\text{perp}}$ always takes on a value between -1 and 1. The derivative is then

$$\frac{\partial k_{\text{geom}}}{\partial \chi} = \frac{0.767}{2\sqrt{2}\sqrt{1 + |\frac{\kappa_{\text{attach}}}{\kappa_{\text{perp}}}|}} \text{sgn}\left(\frac{\kappa_{\text{attach}}}{\kappa_{\text{perp}}}\right) \frac{\frac{\partial \kappa_{\text{attach}}}{\partial \chi}\kappa_{\text{perp}} - \kappa_{\text{attach}}\frac{\partial \kappa_{\text{perp}}}{\partial \chi}}{\kappa_{\text{perp}}^2} \tag{70}$$

3. Sweep Effects

Swept attachment lines result in lower heating rates. In HI-Mach, the correction for sweep is only applied to cylindrical stagnation regions, defined as a region where $|\kappa_{\text{attach}}/\kappa_{\text{perp}}| > 3$. The sweep angle is calculated as

$$\Lambda = |\arccos(|\hat{\boldsymbol{a}}_L \cdot \boldsymbol{v}_\infty|) - \frac{\pi}{2}| \tag{71}$$

where the attachment line direction $\hat{\boldsymbol{a}}_L$ and the freestream direction $\boldsymbol{v}_\infty$ are assumed to be unit-length. The effects of sweep are included in the heat rate by multiplication of a sweep factor

$$k_{\text{sweep}} = \cos^{1.1}\Lambda + 0.05\sin^2\Lambda \tag{72}$$

The derivatives of these quantities are then

$$\frac{\partial \Lambda}{\partial \chi} = \frac{\text{sgn}(\hat{\boldsymbol{a}}_L \cdot \hat{\boldsymbol{v}}_\infty)}{\sqrt{1 - (\hat{\boldsymbol{a}}_L \cdot \hat{\boldsymbol{v}}_\infty)^2}}\left(\frac{\partial \hat{\boldsymbol{a}}_L}{\partial \chi} \cdot \hat{\boldsymbol{v}}_\infty\right) \tag{73}$$

$$\frac{\partial k_{\text{sweep}}}{\partial \chi} = \frac{\partial \Lambda}{\partial \chi}\sin\Lambda\cos\Lambda\left(0.1 - 1.1(\cos\Lambda)^{-0.9}\right). \tag{74}$$

If the effects of sweep are considered, the derivative of stagnation heating in Eqs. (60) and (61) is multiplied by k_{sweep}, and an additional term must be added to the derivative of the stagnation heating to account for sweep effects.

H. Acreage Heating Derivative

The heating to the acreage region is given by

$$\dot{q}_a = \frac{c_f^*}{2}\rho^* u_e\,(h_{aw} - h_w)\,\text{Pr}^{*-2/3} \tag{75}$$

which takes advantage of the simplified formulation for c_f^* in Eq. (56). The derivative, split into two parts, is

$$\left.\frac{\partial \dot{q}_a}{\partial \chi}\right|_{T_w} = \dot{q}_a\left[c_f^{*-1}\left.\frac{\partial c_f^*}{\partial \chi}\right|_{T_w} + \rho^{*-1}\left.\frac{\partial \rho^*}{\partial \chi}\right|_{T_w} + (h_{aw} - h_w)^{-1}\left.\frac{\partial h_{aw}}{\partial \chi}\right|_{T_w} + u_e^{-1}\frac{\partial u_e}{\partial \chi} - \frac{2}{3}\text{Pr}^{*-1}\left.\frac{\partial \text{Pr}^*}{\partial \chi}\right|_{T_w}\right] \tag{76}$$

and

$$\frac{\partial \dot{q}_a}{\partial T_w} = \dot{q}_a\left[c_f^{*-1}\frac{\partial c_f^*}{\partial T_w} + \rho^{*-1}\frac{\partial \rho^*}{\partial T_w} - (h_{aw} - h_w)^{-1}\frac{\partial h_w}{\partial T_w} - \frac{2}{3}\text{Pr}^{*-1}\frac{\partial \text{Pr}^*}{\partial T_w}\right] \tag{77}$$

1. Radiation Equilibrium

The wall conditions currently implemented in HI-Mach are fixed wall temperature and radiation equilibrium wall. For fixed wall temperature, the term $\partial T_w/\partial \chi$ is 0. For a radiation wall equilibrium, the wall temperature is a function of the geometry. The governing equation, reproduced from Eq. (44) of [7], is

$$\epsilon_w \sigma T_w^4 = \dot{q}_w \tag{78}$$

Differentiating this with respect to the geometry yields

$$\frac{\partial \dot{q}_w}{\partial \chi} = 4\epsilon_w \sigma T_w^3 \frac{\partial T_w}{\partial \chi} \tag{79}$$

The derivative of the heat flux to the wall can be expressed as

$$\frac{\partial \dot{q}_w}{\partial \chi} = \left.\frac{\partial \dot{q}_w}{\partial \chi}\right|_{T_w} + \frac{\partial \dot{q}_w}{\partial T_w}\frac{\partial T_w}{\partial \chi} \tag{80}$$

Equations (79) and (80) can be equated to yield

$$\frac{\left.\frac{\partial \dot{q}_w}{\partial \chi}\right|_{T_w}}{4\epsilon_w \sigma T_w^3 - \frac{\partial \dot{q}_w}{\partial T_w}} = \frac{\partial T_w}{\partial \chi} \tag{81}$$

which gives a direct solution for $\partial T_w / \partial \chi$. For a stagnation panel, the quantity $\partial \dot{q}_w / \partial T_w$ is given by Eq. (61) and for an acreage panel it is given by Eq. (77).

I. Curvature Derivatives

Calculating the curvature is a complicated multi-step process. The process is full described in [7]. It involves transforming vectors into a local coordinate system, solving least squares problems on each vertex, then rotating and averaging tensors based on corner angles, solving a rank-2 eigenvalue problem, and then transforming $\mathbb{R}^2$ vectors back into $\mathbb{R}^3$. Each of these steps is a function of vertex position and requires intermediate derivatives. Instead of attempting to directly take sensitivities of the curvature process, central differencing is used to compute the derivatives of the curvature.

The curvature on a panel is a function of the normal vectors of the three vertices which make up the panel. The normal vector of a given vertex is a function of all vertices adjacent to that vertex. Therefore, the curvature on a panel is a function of the three vertices which define the panel, as well as any adjacent vertices. To compute the derivatives of curvature, each of these vertices must be perturbed in 3 directions, and the curvature problem must be solved. This can be expensive, depending on the mesh connectivity.

Thankfully, these derivatives are only necessary on attachment panels. For most geometries, the number of acreage panels vastly outweighs the number of stagnation panels. Table 1 shows the proportion of attachment panels on a variety of geometries.

Geometry Name	Description	Attachment Panel Count	Total Panel Count	Proportion of Attachment Panels (%)
HL-20	Winged lifting re-entry vehicle	711	13546	5.25
GHV	Streamlined low hypersonic aircraft	2096	44624	4.70
HB-2	Blunt nosed, tri-conic flared cone	5	13629	0.037
Sphere-Cone	Blunt nosed, single cone angle	4	58924	0.0068

Table 1 Proportion of attachment line panels across various geometries

More complex geometries with wings, fins, and chines have a higher proportion of attachment panels. Simple blunt-nosed geometries have very low proportions of attachment panels. In general, the more complex the flow pattern across a geometry, the more attachment panels. Nonetheless, even the most complex geometries have less than 5% attachment panels. As a result, it is not prohibitively expensive to compute the derivatives of curvature using central differencing.

HI-Mach implements two methods for mesh curvature calculations, which are described in detail in [7]. The difference-of-normals method only uses 1-ring vertices, while the paraboloid method uses 3-ring vertices. As a result, the curvature when computed with the paraboloid method depends on many more vertices. Therefore, computing derivatives via central differencing for the the difference-of-normals method is much cheaper than the paraboloid method, since less perturbations must be performed.

J. Running Length

The running length is the distance a streamline travels from a stagnation point to a certain point on the vehicle. The running length is used to compute the surface Reynolds number, which is in turn used to compute the convective heat flux to the surface. The running length on panel i is computed by

$$x_i = \sum_{j}^{N} x_j \tag{82}$$

where the sum traverses all the panels from the stagnation point to panel i. In other words, the running length on a certain panel is the sum of discrete running lengths on the preceding panels. The contribution to overall running length from panel j is

$$x_j = |\hat{x}_j - \hat{x}_{j-1}| \tag{83}$$

where $\hat{x}_j$ and $\hat{x}_{j-1}$ are the points where the streamline leaves panels j and $j-1$, respectively. The method for calculating running length is described in detail in [6].

The sensitivity of the total running length is

$$\frac{\partial x}{\partial \chi} = \sum_{i}^{N} \frac{\partial x_i}{\partial \boldsymbol{\chi}} \tag{84}$$

where the sensitivity of each individual running length is given as

$$\frac{\partial x_i}{\partial \chi} = \frac{(\hat{x}_i - \hat{x}_{i-1}) \cdot \left(\frac{\partial \hat{x}_i}{\partial \chi} - \frac{\partial \hat{x}_{i-1}}{\partial \chi}\right)}{|\hat{x}_i - \hat{x}_{i-1}|} \tag{85}$$

The end point of the panel is determined using an intersection between an inner and outer streamline point and the panel edge as described earlier. Thus, the sensitivity of the endpoint of each streamline on a panel is given as

$$\begin{aligned}\frac{\partial \hat{x}_i}{\partial \boldsymbol{\chi}} = &- \frac{\partial \Lambda_j(\hat{x}_{out})}{\partial \boldsymbol{\chi}} \frac{\hat{x}_{in} - \hat{x}_{out}}{\Lambda_j(\hat{x}_{in}) - \Lambda_j(\hat{x}_{out})} \\ &- \Lambda_j(\hat{x}_{out}) \frac{\left(\Lambda_j(\hat{x}_{in}) - \Lambda_j(\hat{x}_{out})\right)\left(\frac{\partial \hat{x}_{in}}{\partial \chi} - \frac{\partial \hat{x}_{out}}{\partial \chi}\right) - \left(\frac{\partial \Lambda_j(\hat{x}_{in})}{\partial \chi} - \frac{\partial \Lambda_j(\hat{x}_{out})}{\partial \chi}\right)(\hat{x}_{in} - \hat{x}_{out})}{\left(\Lambda_j(\hat{x}_{in}) - \Lambda_j(\hat{x}_{out})\right)^2} + \frac{\partial \hat{x}_{out}}{\partial \boldsymbol{\chi}}\end{aligned} \tag{86}$$

where the sensitivities of the barycentric coordinates is calculated by

$$\frac{\partial [\Lambda]}{\partial \boldsymbol{\chi}} = [R]^{-1} \begin{Bmatrix} 0 \\ \frac{\partial x}{\partial \chi} \\ \frac{\partial y}{\partial \chi} \end{Bmatrix} - [R]^{-1} \frac{\partial [R]}{\partial \boldsymbol{\chi}} [R]^{-1} \begin{Bmatrix} 1 \\ x \\ y \end{Bmatrix} \tag{87}$$

The sensitivities of the outer and inner points can be can be calculated by inputting the respective parameters t_{out} and t_{in} into

$$\frac{\partial \hat{x}}{\partial \chi} = \frac{\partial}{\partial \chi} \begin{bmatrix} \xi_1 & \eta_1 \\ \xi_2 & \eta_2 \end{bmatrix} \begin{Bmatrix} B_1 e^{\lambda_1 t} \\ B_2 e^{\lambda_2 t} \end{Bmatrix} + \begin{bmatrix} \xi_1 & \eta_1 \\ \xi_2 & \eta_2 \end{bmatrix} \begin{Bmatrix} \frac{\partial B_1}{\partial \chi} e^{\lambda_1 t} + B_1 \left(\frac{\partial \lambda_1}{\partial \chi} t\right) e^{\lambda_1 t} \\ \frac{\partial B_2}{\partial \chi} e^{\lambda_2 t} + B_2 \left(\frac{\partial \lambda_2}{\partial \chi} t\right) e^{\lambda_2 t} \end{Bmatrix} + \begin{Bmatrix} \frac{\partial x_{cp}}{\partial \chi} \\ \frac{\partial y_{cp}}{\partial \chi} \end{Bmatrix} \tag{88}$$

where

$$\frac{\partial}{\partial \chi} \begin{Bmatrix} B_1 \\ B_2 \end{Bmatrix} = -[E]^{-1} \frac{\partial [E]}{\partial \chi} [E]^{-1} \begin{bmatrix} x^l_{i-1} - x_{cp} \\ y^l_{i-1} - y_{cp} \end{bmatrix} + [E]^{-1} \begin{Bmatrix} \frac{\partial x^l_{i-1}}{\partial \chi} - \frac{\partial x_{cp}}{\partial \chi} \\ \frac{\partial y^l_{i-1}}{\partial \chi} - \frac{\partial y_{cp}}{\partial \chi} \end{Bmatrix} \tag{89}$$

$$\frac{\partial \lambda}{\partial \chi} = \frac{\partial b_1}{\partial \chi} + \frac{\partial c_2}{\partial \chi} \pm \frac{2(b_1 + c_2)(\frac{\partial b_1}{\partial \chi} + \frac{\partial c_2}{\partial \chi}) - 4\left(\frac{\partial b_1}{\partial \chi} c_2 + b_1 \frac{\partial c_2}{\partial \chi} - \frac{\partial b_2}{\partial \chi} c_1 - b_2 \frac{\partial c_1}{\partial \chi}\right)}{4\sqrt{(b_1 + c_2)^2 - 4(b_1 c_2 - b_2 c_1)}} \tag{90}$$

$$\frac{\partial}{\partial \chi}\begin{Bmatrix}\xi_1\\ \xi_2\end{Bmatrix} = \begin{Bmatrix}\frac{\partial \lambda_1}{\partial \chi} - \frac{\partial c_2}{\partial \chi}\\ \frac{\partial c_1}{\partial \chi}\end{Bmatrix} \text{ or } \begin{Bmatrix}\frac{\partial b_2}{\partial \chi}\\ \frac{\partial \lambda_1}{\partial \chi} - \frac{\partial b_1}{\partial \chi}\end{Bmatrix} \tag{91}$$

$$\frac{\partial}{\partial \chi}\begin{Bmatrix}\eta_1\\ \eta_2\end{Bmatrix} = \begin{Bmatrix}\frac{\partial \lambda_2}{\partial \chi} - \frac{\partial c_2}{\partial \chi}\\ \frac{\partial c_1}{\partial \chi}\end{Bmatrix} \text{ or } \begin{Bmatrix}\frac{\partial b_2}{\partial \chi}\\ \frac{\partial \lambda_2}{\partial \chi} - \frac{\partial b_1}{\partial \chi}\end{Bmatrix} \tag{92}$$

$$\frac{\partial x_{cp}}{\partial \chi} = \frac{(b_1c_2 - b_2c_1)\left(\frac{\partial a_2}{\partial \chi}c_1 + a_2\frac{\partial c_1}{\partial \chi} - \frac{\partial a_1}{\partial \chi}c_2 - a_1\frac{\partial c_2}{\partial \chi}\right) - (a_2c_1 - a_1c_2)\left(\frac{\partial b_1}{\partial \chi}c_2 + b_1\frac{\partial c_2}{\partial \chi} - \frac{\partial b_2}{\partial \chi}c_1 - b_2\frac{\partial c_1}{\partial \chi}\right)}{(b_1c_2 - b_2c_1)^2} \tag{93}$$

$$\frac{\partial y_{cp}}{\partial \chi} = \frac{(b_1c_2 - b_2c_1)\left(\frac{\partial a_1}{\partial \chi}b_2 + a_1\frac{\partial b_2}{\partial \chi} - \frac{\partial a_2}{\partial \chi}b_1 - a_2\frac{\partial b_1}{\partial \chi}\right) - (a_1b_2 - a_2b_1)\left(\frac{\partial b_1}{\partial \chi}c_2 + b_1\frac{\partial c_2}{\partial \chi} - \frac{\partial b_2}{\partial \chi}c_1 - b_2\frac{\partial c_1}{\partial \chi}\right)}{(b_1c_2 - b_2c_1)^2} \tag{94}$$

K. Velocity

For each panel on the surface, a linear velocity distribution is defined to allow for tracing of streamlines and attachment line detection. The velocity distribution on the surface of the panel takes the form

$$\begin{Bmatrix}u^l\\ v^l\end{Bmatrix} = \begin{Bmatrix}a_1\\ a_2\end{Bmatrix} + \begin{bmatrix}b_1 & c_1\\ b_2 & c_2\end{bmatrix}\begin{Bmatrix}x^l\\ y^l\end{Bmatrix} \tag{95}$$

where x^l and y^l are the local coordinates in the plane of the panel. The process for determining the coefficients in Eq. ((95)) is described in [6].

The derivative of the coefficients in Eq. ((95)) is given by

$$\frac{\partial\{J\}}{\partial \chi} = \frac{\partial[Q]}{\partial \chi}\{S\} + [Q]\frac{\partial\{S\}}{\partial \chi} \tag{96}$$

where

$$\{J_u\} = \begin{Bmatrix}a_1\\ b_1\\ c_1\end{Bmatrix} \tag{97}$$

$$\{J_v\} = \begin{Bmatrix}a_2\\ b_2\\ c_2\end{Bmatrix} \tag{98}$$

First evaluating $\frac{\partial\{S\}}{\partial \chi}$

$$\frac{\partial\{S_u\}}{\partial \chi} = \begin{Bmatrix}\frac{\partial u_i^l}{\partial \chi}\\ \frac{\partial u_{n_1}^l}{\partial \chi}\\ \frac{\partial u_{n_2}^l}{\partial \chi}\\ \frac{\partial u_{n_3}^l}{\partial \chi}\end{Bmatrix}; \qquad \frac{\partial\{S_v\}}{\partial \chi} = \begin{Bmatrix}\frac{\partial v_i^l}{\partial \chi}\\ \frac{\partial v_{n_1}^l}{\partial \chi}\\ \frac{\partial v_{n_2}^l}{\partial \chi}\\ \frac{\partial v_{n_3}^l}{\partial \chi}\end{Bmatrix} \tag{99}$$

$$\frac{\partial}{\partial \chi}\begin{Bmatrix}u^l\\ v^l\\ w^l\end{Bmatrix} = \frac{\partial}{\partial \chi}[A_{g\to l}]\{V_i\} + [A_{g\to l}]\frac{\partial\{V_i\}}{\partial \chi} \tag{100}$$

$$\frac{\partial V_i}{\partial \chi} = \frac{\partial V_e}{\partial \chi}\frac{\hat{n}_i \times V_\infty \times \hat{n}_i}{|\hat{n}_i \times V_\infty \times \hat{n}_i|} + V_e\left(\frac{(|\hat{n}_i \times V_\infty \times \hat{n}_i|)\frac{\partial}{\partial \chi}(\hat{n}_i \times V_\infty \times \hat{n}_i) - (\hat{n}_i \times V_\infty \times \hat{n}_i)\frac{\partial}{\partial \chi}(|\hat{n}_i \times V_\infty \times \hat{n}_i|)}{|\hat{n}_i \times V_\infty \times \hat{n}_i|^2}\right) \tag{101}$$

$$\frac{\partial}{\partial \chi}\left(\hat{n}_i \times V_\infty \times \hat{n}_i\right) = \left(\frac{\partial \hat{n}_i}{\partial \chi} \times V_\infty \times \hat{n}_i\right) + \left(\hat{n}_i \times V_\infty \times \frac{\partial \hat{n}_i}{\partial \chi}\right) \tag{102}$$

$$\frac{\partial}{\partial \chi}\left(|\hat{n}_i \times V_\infty \times \hat{n}_i|\right) = \frac{\partial}{\partial \chi}\left(\hat{n}_i \times V_\infty \times \hat{n}_i\right) \cdot \frac{\hat{n}_i \times V_\infty \times \hat{n}_i}{|\hat{n}_i \times V_\infty \times \hat{n}_i|} \tag{103}$$

Second, evaluating $\frac{\partial [Q]}{\partial \chi}$

$$\frac{\partial [Q]}{\partial \chi} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ \frac{\partial}{\partial \chi}\left(\left([\Sigma]^T[\Sigma]\right)^{-1}[\Sigma]^T\right) & \begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} \end{bmatrix} \tag{104}$$

$$\frac{\partial}{\partial \chi}\left(\left([\Sigma]^T[\Sigma]\right)^{-1}[\Sigma]^T\right) = \frac{\partial}{\partial \chi}\left(\left([\Sigma]^T[\Sigma]\right)^{-1}\right)[\Sigma]^T + \left([\Sigma]^T[\Sigma]\right)^{-1}\frac{\partial [\Sigma]^T}{\partial \chi} \tag{105}$$

$$\frac{\partial}{\partial \chi}\left(\left([\Sigma]^T[\Sigma]\right)^{-1}\right) = -\left([\Sigma]^T[\Sigma]\right)^{-1}\frac{\partial\left([\Sigma]^T[\Sigma]\right)}{\partial \chi}\left([\Sigma]^T[\Sigma]\right)^{-1} \tag{106}$$

$$\frac{\partial\left([\Sigma]^T[\Sigma]\right)}{\partial \chi} = \frac{\partial [\Sigma]^T}{\partial \chi}[\Sigma] + [\Sigma]^T\frac{\partial [\Sigma]}{\partial \chi} \tag{107}$$

$$\frac{\partial [\Sigma]^T}{\partial \chi} = \left[\frac{\partial [\Sigma]}{\partial \chi}\right]^T \tag{108}$$

L. Attachment Lines

Attachment lines are used to identify where streamlines start on the body and where stagnation heating is applied as opposed to acreage heating. The first attachment line detection method is by an eigenvalue analysis of the linear velocity field. As described in [6], the panel vertices are transformed into a coordinate system where the eigen vectors are orthogonal. If the panel intersects the the X axis, the panel contains an attachment line and the attachment line direction is the eigen vector. The derivative of the attachment line direction is then given by Eq. ((91)).

The second attachment line detection method is through streamline tracing. As described in [6], if a streamline leaves and immediately returns to a panel, the panel is classified as an attachment panel. The attachment line direction is coincident with the edge across which the streamline 'bounces' back and forth. Therefore, the derivative of the attachment line direction is the derivative of the edge unit vector, derived in Eq. (12).

M. Aerodynamic Force Derivatives

1. Force Coefficient Contribution

The total force coefficient is a 3×1 column vector which includes the total force coefficient in x, y, and z, normalized against a reference area.

$$\boldsymbol{C}_F = \sum_{i=1}^{m} \frac{A_i}{A_{\text{ref}}}\left(\hat{\boldsymbol{v}}_A \tau_{w,i} - \hat{\mathbf{n}}_i C_{p,i}\right) \tag{109}$$

The first part of Eq. (109) is the viscous component, and the second part is the inviscid component. The derivative of the total force coefficient with respect to the geometry is obtained with the product rule

$$\frac{\partial \boldsymbol{C}_F}{\partial \boldsymbol{\chi}} = \frac{1}{A_{\text{ref}}}\sum_{i=1}^{m}\left[\frac{\partial A_i}{\partial \boldsymbol{\chi}}\left(\hat{\boldsymbol{v}}_A \tau_{w,i} - \hat{\mathbf{n}}_i C_{p,i}\right) + A_i\left(\frac{\partial \hat{\boldsymbol{v}}_A}{\partial \boldsymbol{\chi}}\tau_{w,i} + \hat{\boldsymbol{v}}_A\frac{\partial \tau_{w,i}}{\partial \boldsymbol{\chi}} - \frac{\partial \hat{\mathbf{n}}_i}{\partial \boldsymbol{\chi}}C_{p,i} - \hat{\mathbf{n}}_i\frac{\partial C_{p,i}}{\partial \boldsymbol{\chi}}\right)\right] \tag{110}$$

2. Lift and Drag

The lift and drag coefficients are often desired as the final result of an aerodynamic analysis. They can be obtained from the total force coefficient with

$$C_L = \boldsymbol{C}_F \cdot \hat{\boldsymbol{n}}_{\text{lift}}\,, \quad C_D = \boldsymbol{C}_F \cdot \hat{\boldsymbol{v}}_\infty \tag{111}$$

where $\hat{\boldsymbol{n}}_{\text{lift}} = \hat{\boldsymbol{n}}_{\text{pitch}} \times \hat{\boldsymbol{n}}_{\text{nose}}$ and $\hat{\boldsymbol{n}}_{\text{pitch}}$ points out the right wing of the aircraft and $\hat{\boldsymbol{n}}_{\text{nose}}$ points out the nose. The derivatives of these with respect to the geometry are then

$$\frac{\partial C_L}{\partial \boldsymbol{\chi}} = \frac{\partial \boldsymbol{C}_F}{\partial \boldsymbol{\chi}} \cdot \hat{\boldsymbol{n}}_{\text{lift}}\,, \quad \frac{\partial C_D}{\partial \boldsymbol{\chi}} = \frac{\partial \boldsymbol{C}_F}{\partial \boldsymbol{\chi}} \cdot \hat{\boldsymbol{v}}_\infty \tag{112}$$

3. Pitching Moment

The pitching moment is given as

$$\boldsymbol{C}_M = \sum_{i=1}^{m} \frac{A_i}{A_{\text{ref}}\, l_{\text{ref}}} \left(\boldsymbol{x}_{\text{centroid}} - \boldsymbol{x}_{CG}\right) \times \left(\hat{\boldsymbol{v}}_A \tau_{w,i} - \hat{\mathbf{n}}_i C_{p,i}\right) \tag{113}$$

The derivative of the pitching moment is obtained as

$$\begin{aligned} \frac{\partial \boldsymbol{C}_M}{\partial \boldsymbol{\chi}} = \frac{1}{A_{\text{ref}}\, l_{\text{ref}}} \sum_{i=1}^{m} \Bigg\{ & \frac{\partial A_i}{\partial \boldsymbol{\chi}} \left[\boldsymbol{x}_{\text{centroid}} - \boldsymbol{x}_{CG}\right] \times \left[\hat{\boldsymbol{v}}_A \tau_{w,i} - \hat{\mathbf{n}}_i C_{p,i}\right] \\ & + A_i \left[\frac{\partial \boldsymbol{x}_{\text{centroid}}}{\partial \boldsymbol{\chi}} \times \left(\hat{\boldsymbol{v}}_A \tau_{w,i} - \hat{\mathbf{n}}_i C_{p,i}\right) + \left(\boldsymbol{x}_{\text{centroid}} - \boldsymbol{x}_{CG}\right) \times \frac{\partial \left(\hat{\boldsymbol{v}}_A \tau_{w,i} - \hat{\mathbf{n}}_i C_{p,i}\right)}{\partial \boldsymbol{\chi}} \right] \Bigg\} \end{aligned} \tag{114}$$

III. Implementation

The derivatives are implemented in HI-Mach alongside the aerodynamic and heating calculations. All of the direct derivatives are tested against a central-difference calculation on a test mesh.

Often, the derivative of a quantity defined on a per-panel basis is desired. For example, the derivative of the pressure coefficient C_p or boundary layer edge viscosity μ_e. In its full form, the derivative for a panel-defined quantity with respect to the geometry is an $n \times 3$ matrix. However, often the panel-defined quantity usually only varies with respect to the 3 vertices that define the panel. The derivatives of each quantity are then stored internally as 9×1 column vectors.

For example, if panel i is defined by vertices j, k, l, and each vertex is defined as (x_j, y_j, z_j), the derivative of the panel boundary layer edge viscosity $\mu_{e,i}$ is stored internally as

$$\frac{\partial \mu_{e,i}}{\partial \boldsymbol{\chi}} = \begin{bmatrix} \frac{\partial \mu_{e,i}}{\partial x_j} & \frac{\partial \mu_{e,i}}{\partial x_k} & \frac{\partial \mu_{e,i}}{\partial x_l} & \frac{\partial \mu_{e,i}}{\partial y_j} & \frac{\partial \mu_{e,i}}{\partial y_k} & \frac{\partial \mu_{e,i}}{\partial y_l} & \frac{\partial \mu_{e,i}}{\partial z_j} & \frac{\partial \mu_{e,i}}{\partial z_k} & \frac{\partial \mu_{e,i}}{\partial z_l} \end{bmatrix}^T \tag{115}$$

This significantly cuts down on the memory cost of the implementation. However, care must be taken in reconstructing the derivative of quantities defined for the whole mesh (such as the force coefficient C_F) so that the correct derivatives with respect to each vertex are stored.

In other cases, the derivatives vary with respect to other vertices. For example, the derivative of heat flux on a panel near the back of the geometry depends on the vertices along the streamline from the stagnation point to that panel. In that case, the derivative is stored as a sparse matrix, using a sparse matrix library developed by Hoch [19].

All implementations of the direct method were tested against a central-difference method to ensure their validity. Central differencing is subject to truncation error and round-off error. Truncation error dominates for large step sizes, and round-off error dominates for small step sizes. To choose the correct central-difference step size, a test was performed evaluating the error between a central difference and direct derivative at various step sizes. The results are shown in Fig. 1.

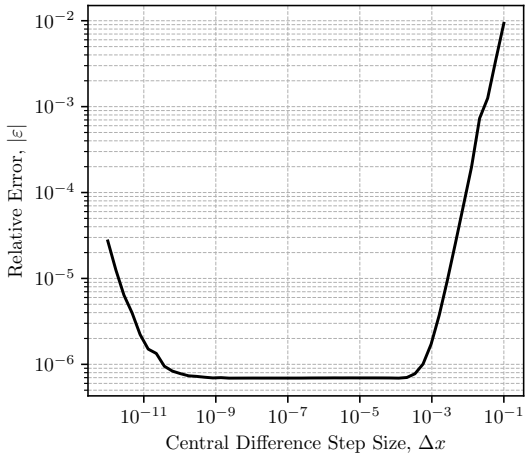


Fig. 1 Absolute error of central-difference method compared to direct method for derivative of lift coefficient

The central difference step size chosen for verification is in the region where neither truncation nor round-off error dominates. The step size was chosen as the cube root of machine precision for an 8-byte floating point number, $h = \sqrt[3]{\epsilon_{\text{machine}}} = 6.055 \times 10^{-6}$.

A. Engineering Sketch Pad Integration

The HI-Mach software is integrated with the Computational Aircraft Prototype Synthesis (CAPS) [20–22] component of the Engineering Sketch Pad (ESP) [23–25] to facilitate sensitivities with respect to geometric BRep (CAD) design parameters [26, 27]. The CAPS framework generates meshes and analysis inputs for HI-Mach, executes the analysis, and processes resultant outputs through an Analysis Interface Module (AIM). In addition, the AIM facilitates completing the chain rule to compute the derivative of functionals with respect to BRep design parameter, $\mathcal{P}$, by completing the chain rule,

$$\frac{\partial C_D}{\partial \mathcal{P}} = \frac{\partial C_D}{\partial \chi} \frac{\partial \chi}{\partial \mathcal{P}} \tag{116}$$

where the vertex sensitivities with respect to BRep design parameters, $\partial \chi / \partial \mathcal{P}$, are computed through ESP [26, 27]. Results from ESP integrated with HI-Mach are included in section IV.C.

IV. Results

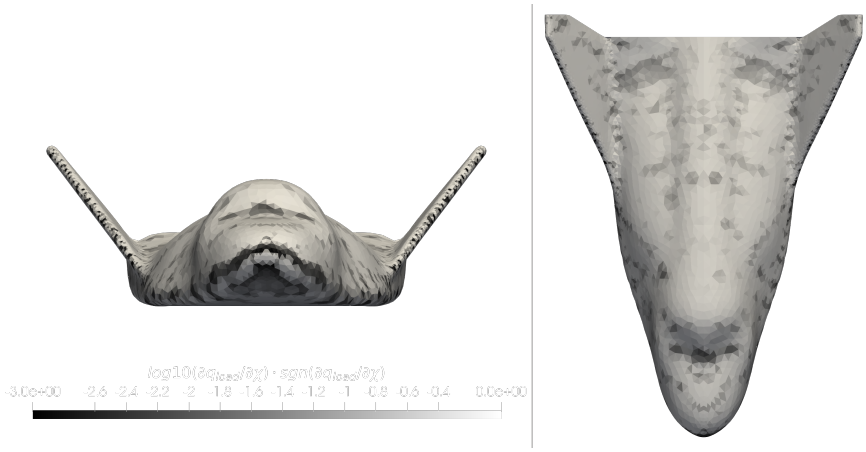
The result of running HI-Mach in sensitivity mode is a set of vectors which represent the derivatives of a given quantity of interest with respect to each of the mesh vertices. This result is most useful as a gradient in a gradient-based optimization routine. For example, the derivative of the lift coefficient with respect to the mesh can be used to optimize a design for maximum lift. The gradients can also be used to constrain an optimization. For example, a hypersonic vehicle can be optimized for maximum lift while constraining drag and maximum wall temperature.

A. Representative Result

A representative result is included below in Fig. 2. The sensitivity of the total heat load to the HL-20 body was computed. To display it, the sensitivity was computed in the direction of the vertex normal vectors. Therefore, a positive value on the surface contour plot indicates that moving the vertex 'out' would result in an increased total heat load. A negative value on the contour plot indicates that moving the vertex 'in' would result in a decreased total heat load.

The data in Fig. 2 is log-scaled. From the figure, the highest magnitude areas are on the nose, chines, and fins. These are attachment areas, subject to higher heating magnitudes than acreage areas. In addition, a high heat rate in an attachment region transmits downstream and increases acreage heating. The result that total heating sensitivity is highest in attachment regions is expected from the theory, and observed in Fig. 2.

(a) Positive component: $\mathrm{sgn}\left(\frac{\partial q_{\mathrm{load}}}{\partial \chi}\right) > 0$. Moving these panels out increases the total heat load.



(b) Negative component: $\mathrm{sgn}\left(\frac{\partial q_{\mathrm{load}}}{\partial \chi}\right) < 0$. Moving these panels out decreases the total heat load.

Fig. 2 Separated derivative of total heat load onto the vehicle. Magnitude is shown using the representation $\mathrm{sgn}\left(\frac{\partial q_{\mathrm{load}}}{\partial \chi}\right) \log_{10}\left|\frac{\partial q_{\mathrm{load}}}{\partial \chi}\right|$.

It can also be seen that pulling in the tip of the nose and pushing out the surrounding panels would decrease the total heat load. The result agrees with the classic Allen & Eggers law, that blunting the nose of a hypersonic vehicle reduces the heating.

B. Runtime Comparison

The purpose of this work is to provide sensitivities of performance quantities such as lift, drag, and temperature with respect to the geometry. The simplest way to calculate these sensitivities is through finite differencing. First, the solution is run without perturbing any points. Then, a point on the geometry is perturbed and the solution is re-run. Three perturbations are required for each point on the geometry for a first-order forward- or backward-difference approach. Six perturbations are required for a central-difference approach. Therefore, $6n$ complete solutions are required to compute the shape sensitivities of a geometry with central differencing. The prohibitive cost of central-differencing is part of the motivation for directly calculating the shape sensitivities.

The runtime of a central-difference method for calculating sensitivities will be compared to the runtime using the approach to calculating sensitivities described in this paper. When running finite-differencing, different approaches can be taken to minimize the run time. The simplest approach would be to run HI-Mach start to finish $6n$ times. However, with that approach the surface mesh is loaded from file every time, which is slow.

Another example would be applying a heuristic and deciding which vertices to central-difference against. For example, when calculating the properties after perturbing a vertex on the upper surface of an aircraft, the bottom surface is not affected, and so it is only necessary to recalculate the top surface.

In making this central difference comparison, the authors decided to time only the calculation period of the code, excluding and I/O. No heuristics were used in the finite-differencing, since these are not known a priori and thus make an unfair comparison.

The timings were performed on spheres and Sears-Haack bodies of varying vertex count. At higher vertex counts, it is prohibitively expensive to run $6n$ cases for the central-differencing. Instead, the code execution was timed and averaged for a smaller number of runs and extrapolated. The run settings which affect the execution time of the code are:

- Wall Temperature: Radiation Equilibrium
- Attachment Line Detection: Eigenvalue and Streamline Tracing
- Boundary Layer Model: Laminar
- Gas Data: Tabulated
- Streamline Density: Sparse

The settings were maintained for both the finite-differencing and direct runs of the code.

The timing comparisons were performed on an i9-12900H with 64Gb of DDR5 memory. Multithreading options exist for HI-Mach both in the base and derivative form. However, to ensure a fair timing comparison, the code was compiled to run on a single thread. The runtime comparisons on a sphere and Sears-Haack body are shown in Fig. 3 and 4

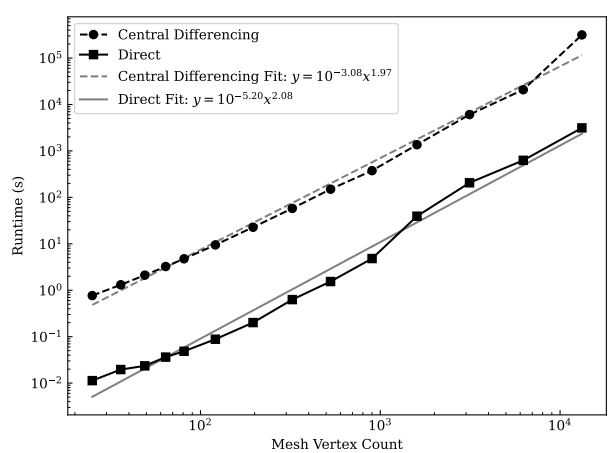


Fig. 3 Runtime comparisons for sphere geometry between central-differencing and direct derivative calculation

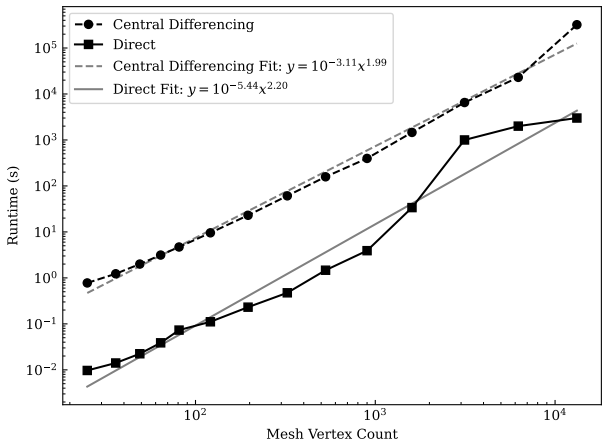


Fig. 4 Runtime comparisons for Sears-Haack geometry between central-differencing and direct derivative calculation

Figures 3 and 4 demonstrate interesting results. The direct method of calculating derivatives is approximately two orders of magnitude faster than central differencing. Based on the scaling, central differencing on large meshes (more than 10,000 mesh vertices) takes more than a day. In comparison, the direct method can execute a 10,000 vertex mesh in less than an hour.

One interesting aspect is the scaling. Both the central differencing and direct methods appear to have approximately $O(n^2)$ scaling. In both methods the most computationally expensive aspect is computing the streamlines (and the sensitivity of the streamlines). In Navier-Stokes CFD, the adjoint (or sensitivity) can be computed for the same cost as the base solution. Therefore, computing the sensitivities with the adjoint method in CFD scales better than computing them with finite-differencing. However, computing the sensitivity in this panel method scales the same as finite-differencing, offset by a finite cost. This agrees with the scaling results found by Hoch in a supersonic panel code [19].

Departures from $O(n^2)$ are observed. Some of the deviation is likely due to memory effects. The direct derivative method stores multiple arrays of size $m \times n$, so its memory scaling is approximately $O(n^2)$. RAM access speed is nonlinear. Large slowdowns were observed during the scaling study and attributed to large memory requirements.

C. Engineering Sketch Pad

An Analysis Interface Module (AIM) was developed between HI-Mach and ESP. A Sphere-Cone geometry is used to demonstrate the coupling for sensitivity calculations. The sphere-cone is a union of a spherical nose with a conical frustum. It has five geometric shape parameters given in Table 2. The R parameter defines the radius of the sphere at the nodes, H is the diameter of the trailing base, and L is the length between the center of the sphere and the circular base. The final two parameters, α_g and β_g, are simple rotations of the geometry without any other shape changes. These are equivalent to changing the analysis input angles of attach and side slip angles. However, they serve the purpose of demonstrating shape sensitivities due to simple rotations.

The results of C_D geometric parameter sweeps and sensitivities are shown in Fig. 6. For each sub-figure, the respective geometric parameter is swept over some range, and for each parameter value the geometry is updated, re-meshed, and both C_D and $\frac{\partial C_D}{\partial \chi}$ are computed via HI-Mach, and $\frac{\partial \chi}{\partial \mathcal{P}}$ is computed using ESP and the complete sensitivity, $\frac{\partial C_D}{\partial \mathcal{P}}$, is computed using the AIM. The C_D values are illustrated by the blue curve associated with the left axes, and the sensitivity values, $\frac{\partial C_D}{\partial \mathcal{P}}$, are illustrated with the red curves associated with the right axes. In addition, a central difference of the C_D values, $\frac{\Delta C_D}{\Delta \mathcal{P}}$, is shown as a black curve. The comparison of the black and red curves provides a qualitative comparison of the accuracy of the analytic sensitivity calculation over a range of each geometric parameter. As evident from Fig. 6, the analytic and finite difference sensitivities agree well across the range of all geometric parameters. This stands as verification that HI-Mach and ESP are correctly coupled for computation of sensitivities with respect to design parameters.

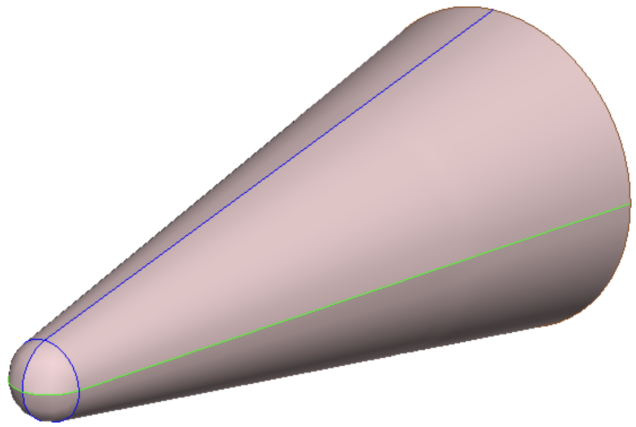
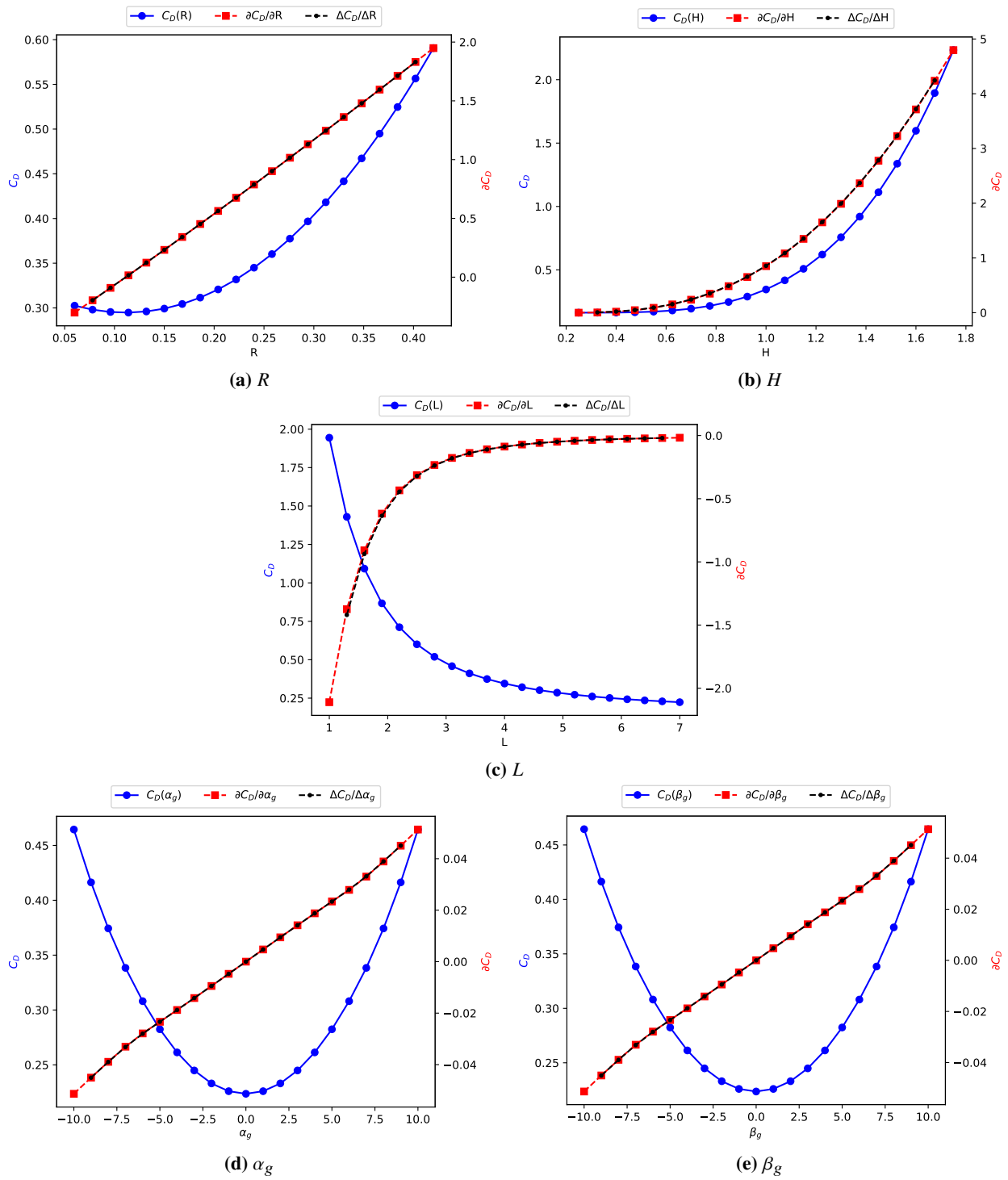


Fig. 5 ESP sphere-cone body

Table 2 ESP Sphere-Cone Geometric Parameters

Design Parameter Description	
R	Radius of the sphere
H	Radius of the circular base
L	Length between the center of the sphere and the circular base
α_g	Geometric "angle of attack"
β_g	Geometric "side slip angle"



(a) R **(b)** H

(c) L

(d) α_g **(e)** β_g

Fig. 6 ESP sphere-cone drag sensitivities w.r.t. geometry parameters

V. Conclusion

Calculation of sensitivities of quantities of interest such as heat flux, lift, and drag are useful for optimizing a hypersonic design. These sensitivities can be derived analytically for a hypersonic impact method. In [6] and [7], the authors present a hypersonic impact method for calculation of aerothermodynamic quantities. An approach to obtain the

sensitivities of that method are explored and derived. Direct analytical sensitivities are obtained where possible. For difficult calculations such as curvature, an alternative approach to obtaining the sensitivities is described.

Results of the present calculations demonstrate that the sensitivites are of computational complexity $O(n^2)$. Compared to finite-differencing, the method presented in this work is approximately two orders of magnitude faster. The sensitivity of a quantity of interest with respect to a mesh can be converted to the sensitivity with respect to a design parameter. This conversion is performed by ESP, which is a widely-used tool for vehicle design. Coupling the sensitivity mode in HI-Mach with ESP allows users a flexible and convenient framework for optimizing vehicle designs. The methods and approaches presented here allow for agile development and optimization of hypersonic vehicles.

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References

[1] Kline, H., and Alonso, J., "Adjoint of generalized outflow-based functionals applied to hypersonic inlet design," *AIAA Journal*, Vol. 55, No. 11, 2017, pp. 3903–3915.

[2] Fu, J., Qu, F., Liu, Q., Sun, D., and Bai, J., "Multi-objective aerodynamic optimization of two-dimensional hypersonic forebody-inlet based on the heuristic algorithm," *Aerospace Science and Technology*, Vol. 123, 2022, p. 107470.

[3] Damm, K. A., Gollan, R. J., Jacobs, P. A., Smart, M. K., Lee, S., Kim, E., and Kim, C., "Discrete adjoint optimization of a hypersonic inlet," *Aiaa Journal*, Vol. 58, No. 6, 2020, pp. 2621–2634.

[4] Eyi, S., Hanquist, K. M., and Boyd, I. D., "Aerothermodynamic design optimization of hypersonic vehicles," *Journal of Thermophysics and Heat Transfer*, Vol. 33, No. 2, 2019, pp. 392–406.

[5] Cui, K., and Yang, G.-W., "Shape optimization for hypersonic arc-wing missiles," *Journal of Spacecraft and Rockets*, Vol. 47, No. 4, 2010, pp. 694–700.

[6] Goates, J. K., Freeman, L. B., and Hunsaker, D. F., "Theory and Implementation of a Rapid Hypersonic Impact Method (HI-Mach) Part I: Aerodynamics," *AIAA SciTech 2025 Forum*, AIAA, Orlando, FL, 2025. https://doi.org/10.2514/6.2025-2263.

[7] Freeman, L. B., Goates, J. K., and Hunsaker, D. F., "Theory and Implementation of a Rapid Hypersonic Impact Method (HI-Mach) Part II: Convective Heating," *AIAA SciTech 2025 Forum*, AIAA, Orlando, FL, 2025. https://doi.org/10.2514/6.2025-2264.

[8] Kinney, D., "Aero-Thermodynamics for Conceptual Design," *42th AIAA Aerospace Sciences Meeting and Exhibit*, 2004. https://doi.org/10.2514/6.2004-31.

[9] Kinney, D., "Aerodynamic shape optimization of hypersonic vehicles," *44th AIAA aerospace sciences meeting and exhibit*, 2004, p. 239.

[10] Hodson, J., "Numerical Analysis and Spanwise Shape Optimization for Finite Wings of Arbitrary Aspect Ratio," Ph.D. thesis, 2019.

[11] Hoch, N., Bolander, C., and Hunsaker, D., "Application of the Adjoint Method to an Unstructured Subsonic/Supersonic Panel Method," *Journal of Aircraft*, 2025, pp. 1–15.

[12] Dannenhoffer, J., "An Overview of the Engineering Sketch Pad," *AIAA SCITECH 2024 Forum*, 2024, p. 1315.

[13] Peter, J. E., and Dwight, R. P., "Numerical sensitivity analysis for aerodynamic optimization: A survey of approaches," *Computers & Fluids*, Vol. 39, No. 3, 2010, pp. 373–391. https://doi.org/10.1016/j.compfluid.2009.09.013.

[14] Martins, J. R., and Hwang, J. T., "Review and unification of methods for computing derivatives of multidisciplinary computational models," *AIAA journal*, Vol. 51, No. 11, 2013, pp. 2582–2599. https://doi.org/10.2514/1.J052184.

[15] Kenway, G. K., Mader, C. A., He, P., and Martins, J. R., "Effective adjoint approaches for computational fluid dynamics," *Progress in Aerospace Sciences*, Vol. 110, 2019, p. 100542. https://doi.org/10.1016/j.paerosci.2019.05.002.

[16] Martins, J. R., Sturdza, P., and Alonso, J. J., "The complex-step derivative approximation," *ACM Transactions on Mathematical Software (TOMS)*, Vol. 29, No. 3, 2003, pp. 245–262.

[17] Müller, J.-D., and Cusdin, P., "On the performance of discrete adjoint CFD codes using automatic differentiation," *International journal for numerical methods in fluids*, Vol. 47, No. 8-9, 2005, pp. 939–945.

[18] Giles, M. B., and Pierce, N. A., "An Introduction to the Adjoint Approach to Design," *Flow, Turbulence and Combustion*, Vol. 65, 2000, pp. 393–415. https://doi.org/10.1023/A:1011430410075.

[19] Hoch, N. H., "Calculation of Aerodynamic Surface Sensitivities Using the Adjoint Method in a Subsonic/Supersonic Panel Code," Master's thesis, Utah State University, 2025. https://doi.org/10.26076/41db-0522, URL https://digitalcommons.usu.edu/etd2023/425, all Graduate Theses and Dissertations, Fall 2023 to Present, No. 425.

[20] Alyanak, E., Durscher, R., Haimes, R., John F. Dannenhoffer, I., Bhagat, N., and Allison, D., "Multi-fidelity Geometry-centric Multi-disciplinary Analysis for Design," AIAA 2016-4007, June 2016. https://doi.org/10.2514/6.2016-4007.

[21] Durscher, R. J., and Reedy, D., "pyCAPS: A Python Interface to the Computational Aircraft Prototype Syntheses," AIAA 2019-2226, ???? https://doi.org/10.2514/6.2019-2226.

[22] Bryson, D. E., and Haimes, R., "Toward the Realization of a Highly Integrated, Multidisciplinary, Multifidelity Design Environment," AIAA 2019-2225, January 2019. https://doi.org/10.2514/6.2019-2225.

[23] Haimes, R., and Dannenhoffer, J., "The Engineering Sketch Pad: A Solid-Modeling, Feature-Based, Web-Enabled System for Building Parametric Geometry," AIAA 2013–3073, 2013. https://doi.org/10.2514/6.2013-3073.

[24] Dannenhoffer, J., and Haimes, R., "An Integrated Design Environment for the Engineering Sketch Pad," AIAA 2023-0550, 2023. https://doi.org/10.2514/6.2023-0550.

[25] Dannenhoffer, J., "An Overview of the Engineering Sketch Pad," AIAA 2024-1315, 2024. https://doi.org/10.2514/6.2024-1315.

[26] Dannenhoffer, J. F., "OpenCSM: An Open-Source Constructive Solid Modeler for MDAO," AIAA 2013-0701, January 2013. https://doi.org/10.2514/6.2013-701.

[27] Dannenhoffer, J., and Haimes, R., "Design Sensitivity Calculations Directly on CAD-based Geometry," AIAA 2015-1370, 2015. https://doi.org/10.2514/6.2015-1370.